

교육용 자료 안내

본 자료는 Darun Color Institute의 색채 교육 과정을 위해 제공되는 교육용 문서입니다. 학생들이 안전한 재료 취급을 학습하기 위해 사용된 원료의 물리화학적 특성과 안전 정보를 공개합니다.

다만, 이 자료에 포함된 정보는 교육 목적으로만 활용되어야 하며, 상업적 목적의 제품 제조 및 판매는 엄격히 금지됩니다.

화장품 제조업 허가를 취득한 자가 독립적으로 제품을 개발·판매할 경우, 다음을 반드시 준수해야 합니다:

- 신뢰할 수 있는 화장품 원료 전문 제조사로부터 원료를 직접 확보
- 해당 제조사로부터 최신 MSDS 및 시험성적서(CoA, 안정성 시험 등)를 개별 요청
- 식품의약품안전처 규정에 따른 원료 검증 및 제품 신고

이는 소비자 안전과 법적 책임을 보장하기 위한 필수 절차입니다.

Educational Material Notice

This document is provided by Darun Color Institute as an educational resource for our color science training programs. To support student learning in safe material handling practices, we disclose the physical, chemical, and safety characteristics of the raw materials used in our formulations.

However, all information contained in this material is strictly for educational purposes only. Any commercial use for product manufacturing or sales is strictly prohibited.

Should you obtain cosmetic manufacturing authorization and independently develop products for commercial purposes, you must:

- Source raw materials directly from reliable cosmetic ingredient manufacturers
- Request current MSDS and certificates of analysis (CoA) individually from the respective manufacturers
- Verify all raw materials and complete product registration in compliance with Korea Food and Drug Administration (KFDA) regulations

These procedures are mandatory requirements to ensure consumer safety and fulfill legal responsibilities.

Material Safety Data Sheet (MSDS)

1. Chemical Product and Company Identification

Item	Information
Product Name	Educational Color Z-Brown
Recommended Use	Cosmetic raw material
INCI Name	Iron Oxides (CI 77499, CI 77492), Iron Hydroxide Oxide, Water, Cyclomethicone, Caprylic/Capric Triglyceride, Dipropylene Glycol, Glycerin, Propanediol, Cetyl PEG/PPG-10/1 Dimethicone, 1,2-Hexanediol, Sodium Hyaluronate, Butylene Glycol, Distearidimonium Hectorite, Allantoin, Macadamia Ternifolia Seed Oil, DL-Panthenol, Sorbitan Olivatate, Ethylhexylglycerin

2. Hazards Identification

Category	Details
GHS Classification	Specific Target Organ Toxicity (Single Exposure) - Category 3 (Respiratory irritation)
Hazard Statements	H335: May cause respiratory irritation
Precautionary Statements	P261: Avoid inhalation of dust, fumes, gas, mist, vapor, spray P271: Use only outdoors or in well-ventilated areas P304+P340: If inhaled: Move to fresh air and rest in comfortable position P312: Seek medical advice if you feel unwell P403+P233: Store in well-ventilated place. Keep container tightly closed P405: Store in locked storage P501: Dispose of contents/container according to regulations

3. Composition/Information on Ingredients

**Composition ratios are provided as approximate ranges for educational purposes*

Ingredient Name	CAS No.	Approximate Content (%)
Iron Oxide (CI 77499)	1309-37-1	5~12.5%
Black Iron Oxide (CI 77492)	1317-61-9	0.5~2.5%
Iron Hydroxide Oxide	20344-49-4	35~40%
Water	7732-18-5	17.5~32.5%
Cyclomethicone	69430-24-6	2.5~10%
Caprylic/Capric Triglyceride	65381-09-1	2.5~5%
Dipropylene Glycol	110-98-5	2.5~5%
Glycerin	56-81-5	0.5~2.5%
Propanediol	504-63-2	0.5~2.5%

Cetyl PEG/PPG-10/1 Dimethicone	6920-22-5	0.5~2.5%
1,2-Hexanediol	Not specified	1%
Sodium Hyaluronate	9067-32-7	0.5~1%
Butylene Glycol	107-88-0	0.5~1%
Disteardimonium Hectorite	97280-96-1	0.5~1%
Allantoin	97-59-6	0.5~1%
Macadamia Ternifolia Seed Oil	438545-25-6	0.5~1%
DL-Panthenol	16485-10-2	0.5~1%
Sorbitan Olivat	223706-40-9	0.5~1%
Ethylhexylglycerin	70445-33-9	0.5~1%

4. First Aid Measures

Route of Exposure	First Aid Procedure
Eye Contact	Rinse with water for at least 15-20 minutes. Seek medical attention if irritation persists.
Skin Contact	Wash with soap and water for at least 20 minutes. Remove contaminated clothing and shoes before reuse. Seek medical attention if irritation persists.
Inhalation	Move to fresh air immediately. If breathing is difficult, seek medical attention. If coughing or other symptoms occur, seek medical examination.
Ingestion	Rinse mouth with water. Do not induce vomiting. Seek medical attention immediately for large amounts.
Medical Attention	Ensure medical personnel are aware of substance exposure and take appropriate protective measures.

5. Fire-Fighting Measures

Item	Information
Suitable Extinguishing Media	Dry powder, carbon dioxide, water spray, alcohol foam
Specific Hazards	Container may explode when heated. Irritating and toxic gases may be generated during combustion.
Special Protective Equipment	Move container from fire area if safe. Cool containers with water spray even after extinguishing.
Unsuitable Media	High-pressure water streams

6. Accidental Release Measures

Action	Description
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Prevention	Remove all ignition sources. Prevent dust formation. Ventilate contaminated area.
Handling Spills	For small spills: Absorb with sand or non-combustible material and place in container For large spills: Create barriers to prevent spreading. Use clean shovel to place in container.
Environmental Protection	Prevent entry into waterways, sewers, basements, and confined spaces. For powder spills, cover with plastic sheet to prevent diffusion.

7. Handling and Storage

Category	Requirements
Handling	Handle carefully following safety procedures. Use full or local exhaust ventilation. Wash thoroughly after handling. Avoid inhalation.
Storage	Store in cool, dry, and well-ventilated place. Keep container tightly closed. Store away from heat, ignition sources, and incompatible materials.

8. Exposure Controls/Personal Protection

Control Type	Requirement
Engineering Controls	Use local exhaust ventilation or process isolation to maintain air below exposure limits
Respiratory Protection	Use KOSHA-certified respiratory protection appropriate for particulate matter exposure
Eye Protection	Wear safety glasses or goggles. Install emergency eyewash facilities near work area
Hand Protection	Wear appropriate chemical-resistant protective gloves
Body Protection	Wear appropriate chemical-resistant protective clothing
Facilities	Install emergency eyewash and shower facilities in work area

9. Physical and Chemical Properties

Property	Value
Appearance	Cream to paste-like, brown colored semi-solid
Odor	Slight characteristic odor
Viscosity	No data available
pH	No data available
Solubility	Partially soluble; water component soluble
Specific Gravity	No data available

10. Stability and Reactivity

Item	Information
Chemical Stability	Stable at room temperature and normal pressure
Conditions to Avoid	Heat, sparks, flames, and other ignition sources
Incompatible Materials	Combustible materials, oxidizing agents
Hazardous Decomposition	Container may explode when heated. Irritating and toxic gases may be generated during combustion.

11. Toxicological Information

Effect	Description
Acute Toxicity (Oral)	LD50 > 5000 mg/kg (Rat) - Iron Oxide component
Acute Toxicity (Inhalation)	Dust LC50: 5.05 mg/L 4hr (Rat) - Iron Oxide component
Skin Irritation	Minimal irritation potential
Eye Irritation	Minimal irritation potential
Respiratory Irritation (Single Exposure)	May cause respiratory irritation from iron oxide dust component
Long-term Exposure Effects	Prolonged exposure to iron oxide dust may cause respiratory effects
Carcinogenicity	Iron Oxide Group A4 (ACGIH) - Suspected human carcinogen (inhalation hazard for fine particles)
Reproductive Toxicity	No data available
Aspiration Hazard	No data available

12. Ecological Information

Category	Information
Fish Toxicity	LC96hr $\geq$ 50000 mg/L (Iron Oxide component)
Crustacean Toxicity	EC50 > 100 mg/L 48hr (Iron Oxide component)
Bird Toxicity	No data available
Persistency and Degradability	No data available
Bio-condensability	No data available
Soil Mobility	No data available

13. Disposal Considerations

Item	Information
Disposal Method	Dispose of contents and container in accordance with local waste management regulations
Special Precautions	For iron oxide component: Neutralization, hydrolysis, oxidation,

	reduction, incineration, or solidification may be appropriate
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14. Transport Information

Item	Information
UN Number	Not classified as hazardous for transport
Proper Shipping Name	Not applicable
Transport Hazard Class	Not applicable
Packing Group	Not applicable
Marine Pollutant	Not applicable

15. Regulatory Information

Item	Information
Regulatory Basis	Prepared in accordance with Korean Occupational Safety and Health Act and relevant regulations

16. Other Information

Item	Information
Disclaimer	The information contained in this document is based on accurate data. However, no guarantee is provided regarding the accuracy of results or values obtained when using this material for specific purposes or mixing with other substances. The company assumes no responsibility for any personal losses or property damage to buyers or third parties.
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